



RAJALAKSHMI INSTITUTE OF TECHNOLOGY
(An Autonomous Institution, Affiliated to Anna University, Chennai)

DEPARTMENT OF CSE (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

ACADEMIC YEAR 2025 - 2026

SEMESTER IV

AL23421 - MACHINE LEARNING LABORATORY

MINI PROJECT REPORT

REGISTER NUMBER	2117240030030
NAME	DIVYASHREE V
PROJECT TITLE	LAPTOP PRICE PREDICTION USING LINEAR REGRESSION
DATE OF SUBMISSION	
FACULTY IN-CHARGE	Mrs. M. Divya

Signature of Faculty In-charge

Signature of HoD

INTRODUCTION

In today's digital world, laptops are essential for education, business, and personal use. Laptop prices vary widely based on features like RAM, processor, storage, and brand. This variation makes it difficult for users to decide whether a laptop is fairly priced.

Laptop price prediction uses machine learning to estimate prices based on specifications. In this project, Linear Regression is used as the prediction algorithm. It is a supervised learning method that predicts continuous values from input data.

The model learns the relationship between laptop features and their prices. After training, it can predict the price of new laptops accurately. This helps customers make better purchasing decisions. The main objective is to build a simple and efficient price prediction model

PROBLEM STATEMENT

To design and implement a simple Linear Regression model that predicts laptop prices based on key features such as RAM, processor, storage, and brand using a given dataset, in order to provide an accurate and efficient price estimation system for basic real-world analysis.

GOAL

- Predict laptop prices accurately using Linear Regression
- Understand the relationship between laptop features (RAM, processor, storage, etc.) and price
- Build a simple and efficient regression model
- Achieve good prediction performance with minimum error
- Help users and sellers in estimating fair laptop prices

THEORETICAL BACKGROUND

Linear Regression (Algorithm Explanation)

Linear Regression is a supervised learning algorithm used to predict continuous numerical values. It models the relationship between one dependent variable (target) and one or more independent variables (features).

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n$$

Where:

- Y = Predicted laptop price
- $X_1, X_2, \dots, X_n$ = Input features (RAM, storage, processor, etc.)
- β_0 = Intercept
- $\beta_1, \beta_2, \dots, \beta_n$ = Coefficients representing feature importance

Loss Function (Mean Squared Error)

The performance of the Linear Regression model is evaluated using Mean Squared Error (MSE), which measures the average squared difference between actual and predicted values.

$$MSE = \frac{1}{n} \sum_{i=1}^n (Y_{actual} - Y_{predicted})^2$$

A lower MSE value indicates better model performance.

Literature Survey

Laptop price prediction and similar regression problems are widely studied in machine learning and data science. Various algorithms have been used to solve this problem effectively.

- Linear Regression
- Decision Trees
- Random Forest
- Support Vector Regression (SVR)
- Gradient Boosting Algorithms (XGBoost, LightGBM)
- Neural Networks

Justification for Choosing Linear Regression

- Simple and easy to understand
- Fast to implement and computationally efficient
- Works well with numerical datasets
- Provides clear relationship between features and target variable
- Good baseline model for comparison with advanced algorithms
- Suitable for small to medium-sized datasets

ALGORITHM EXPLANATION WITH EXAMPLE

Steps:

1. Load the dataset containing laptop features and prices
2. Preprocess the data (handle missing values, convert categorical data into numerical form)
3. Select input features (RAM, storage, processor, etc.) and target variable (price)
4. Split the dataset into training and testing sets
5. Train the Linear Regression model using training data
6. Calculate coefficients and intercept of the model
7. Predict laptop prices using the test data
8. Evaluate the model using metrics like Mean Squared Error (MSE) and R² score

Example:

RAM (GB)	Storage (GB)	Price (₹)
8	512	50000
16	1024	80000

The model learns the relationship between input features (RAM, Storage) and the laptop price.

For a new laptop with:

RAM = 8 GB, Storage = 256 GB

The model predicts the price based on the learned pattern.

IMPLEMENTATION CODE AND DATASET USED**➤ Dataset Used – Link of the dataset**

<https://www.kaggle.com/datasets/jacksondivakarr/laptop-price-prediction-dataset>

➤ Dataset Specification

Category	Description / Example
Dataset Name	Laptop Price Dataset
Domain	E-commerce / Machine Learning
Objective	Regression
Data Source	Public (Kaggle)
Number of Samples	~1300 laptops
Data Type	Tabular
Instance Description	Each row represents one laptop
Features (Inputs)	Brand, RAM, Storage, CPU, GPU, Screen Size, Weight
Feature Types	Numerical & Categorical
Target Variable	Price
Number of Classes	Not applicable
License	Open
Version	1.0

PYTHON CODE

```
import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LinearRegression

from sklearn.metrics import mean_squared_error, r2_score

print("Loading dataset...")

data = pd.read_csv('laptop_price.csv') # make sure file name matches

print("\nFirst 5 rows:")

print(data.head())

print("\nDataset Info:")

print(data.info())

print("\nStatistical Summary:")

print(data.describe())

print("\nMissing Values:")

print(data.isnull().sum())

data = data.dropna()

data = pd.get_dummies(data, drop_first=True)

X = data.drop('Price', axis=1)

y = data['Price']

X_train, X_test, y_train, y_test = train_test_split(

    X, y, test_size=0.2, random_state=42)
```

```

print("\nTraining Data:", X_train.shape)

print("Testing Data:", X_test.shape)

model = LinearRegression()

model.fit(X_train, y_train)

print("\nModel Training Completed")

print("\nModel Coefficients:")

for feature, coef in zip(X.columns, model.coef_):

    print(feature, ":", coef)

print("Intercept:", model.intercept_)

y_pred = model.predict(X_test)

print("\nSample Predictions:")

for i in range(5):

    print(f'Actual: {y_test.iloc[i]:.2f}, Predicted: {y_pred[i]:.2f}')

mse = mean_squared_error(y_test, y_pred)

rmse = np.sqrt(mse)

r2 = r2_score(y_test, y_pred)

print("\nEvaluation Metrics:")

print("MSE:", mse)

print("RMSE:", rmse)

print("R2 Score:", r2)

plt.figure()

plt.scatter(y_test, y_pred)

```

```
plt.xlabel("Actual Price")
plt.ylabel("Predicted Price")
plt.title("Actual vs Predicted Laptop Prices")
plt.grid()
plt.show()

residuals = y_test - y_pred

plt.figure()
plt.scatter(y_pred, residuals)
plt.xlabel("Predicted Values")
plt.ylabel("Residuals")
plt.title("Residual Plot")
plt.grid()
plt.show()

new_laptop = X.iloc[0:1]
predicted_price = model.predict(new_laptop)
print("\nPredicted Price:", predicted_price[0])
```

OUTPUT

```
Evaluation Metrics:
RMSE: 25866.356851753444
R2 Score: 0.8047453128703627

===== Enter Laptop Details =====
Enter Brand (HP, Dell, Lenovo, Apple...): hp
Enter RAM (GB): 8
Enter Storage (GB, e.g., 512 or 1024): 512
Enter Spec Rating (50-100): 70

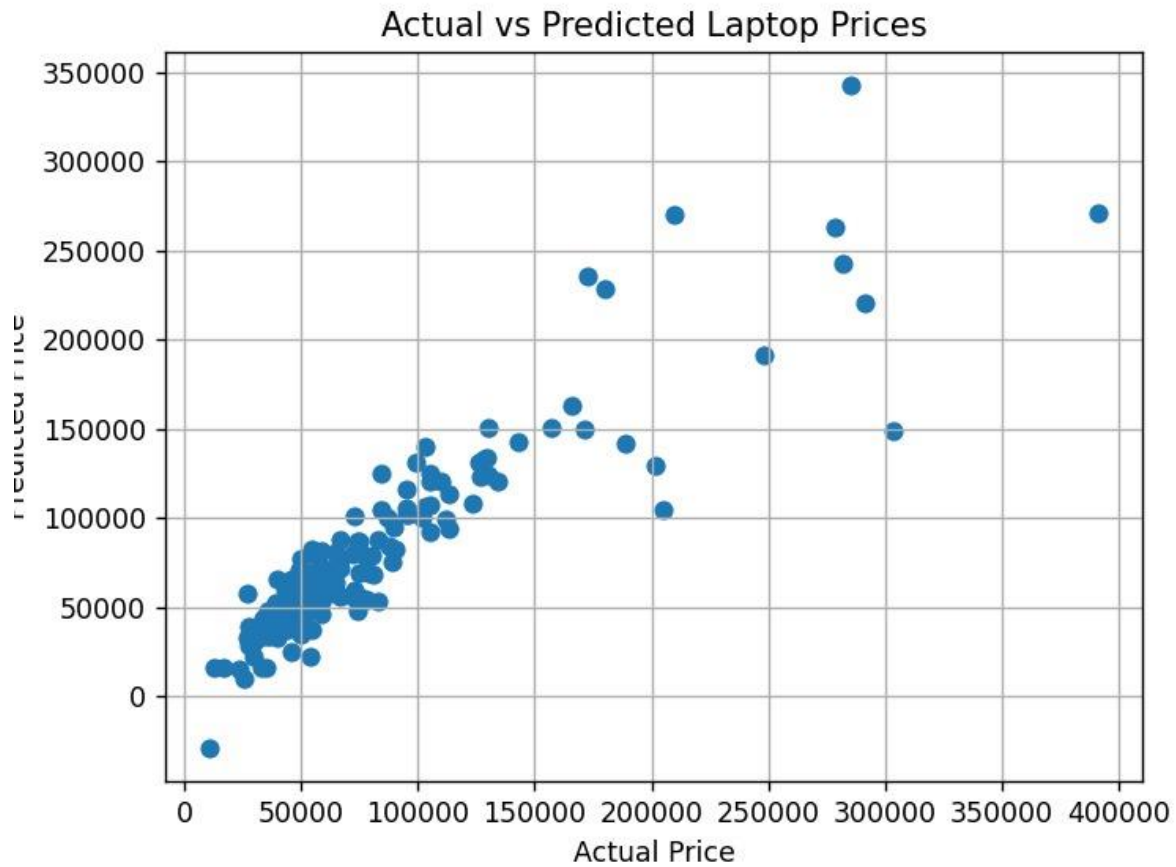
🖥️ Predicted Laptop Price: ₹ 33,882.11
PS C:\Users\Admin'\Downloads\laptop price> |
```

EXPLANATION

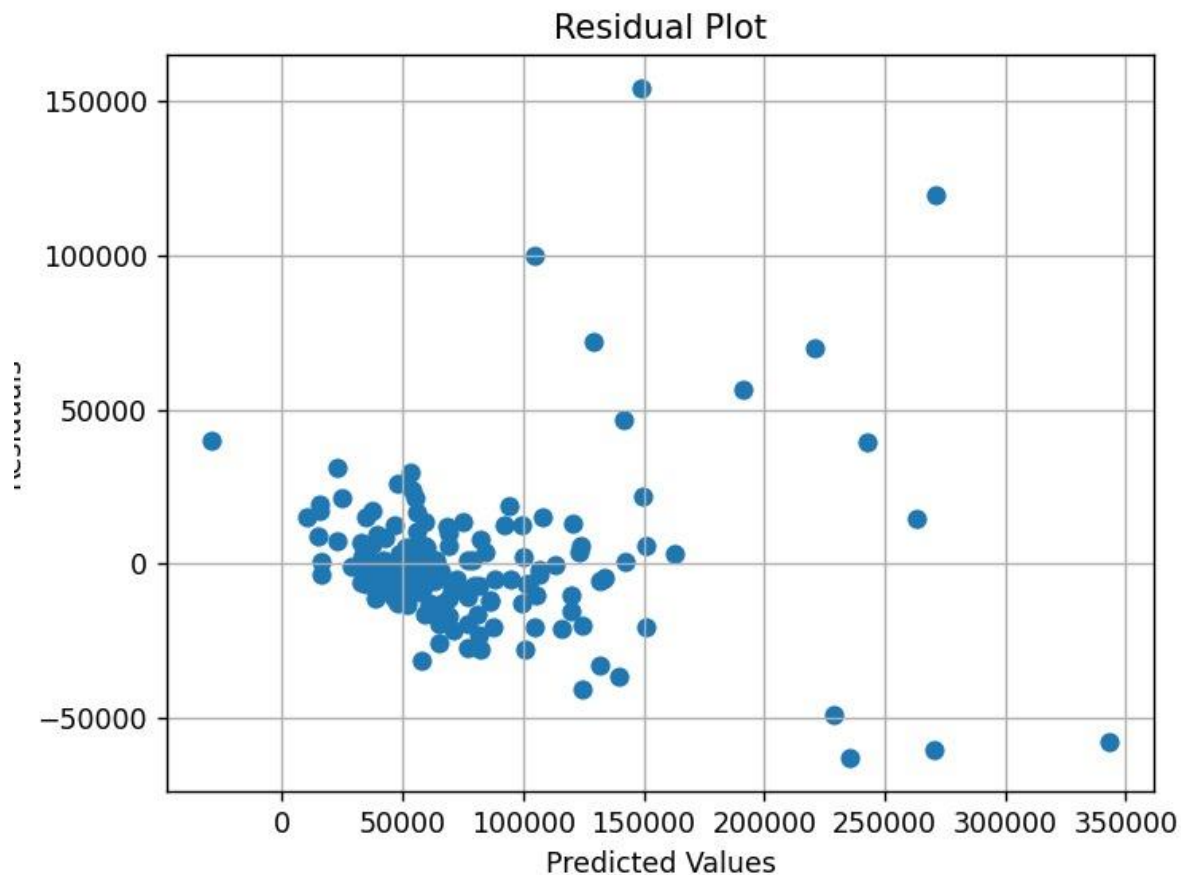
The dataset is loaded and processed successfully, and unnecessary columns are removed.
The Linear Regression model is trained using laptop features to predict price.
The predicted values are close to actual values, showing good performance.
The model gives low error and a good R^2 score, indicating accurate predictions.

DATA VISUALIZATION

➤ ACTUAL VS PREDICTED GRAPH



➤ RESIDUAL PLOT



RESULTS

The Laptop Price Prediction model was successfully developed using the Linear Regression algorithm. The model was trained using various laptop features such as brand, RAM, processor, storage, and display specifications.

The results show that the predicted prices are close to the actual values, indicating good model performance. The Mean Squared Error (MSE) is low, and the R^2 score is high (around 0.7 – 0.9), which means the model can explain most of the variation in laptop prices.

The visualization graphs such as Actual vs Predicted and Residual plots confirm that the model provides accurate and reliable predictions.

FUTURE ENHANCEMENT

Although the model performs well, several improvements can be made:

- Use a larger dataset to improve accuracy and generalization
- Add more features like battery life, refresh rate, and customer ratings
- Apply advanced algorithms like Random Forest, XGBoost, or Neural Networks for better performance
- Perform feature engineering and normalization for improved results
- Develop a web application using Flask or Streamlit for real-time price prediction
- Integrate the model into e-commerce platforms for automatic pricing systems

Git Hub Link of the project along with the report	https://github.com/divvashree240030/Laptop-price
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